# Source Code and Data Description

Evaluating and Reducing Hallucinations in LLM-Based Medical Report Summarization

Srilaya Ponangi · COMP 698C · Fall 2026 · California State University, Northridge · November 10, 2026

## 1. Repository at a glance

| Item | Value |
|---|---|
| Source | https://github.com/slay-a/medical-hallucination-eval (public, branch main) |
| Commit | faa48a8 (2026-11-10) |
| Type | Python 3 research codebase; Bash helper scripts; no compiled components |
| Size | code 809 KB; result artifacts 6.4 MB |
| Language and runtime | Python 3.9.6 in a virtual environment (.venv) for the pipeline; Python 3.13 in .venv-llm for the mlx-lm model server; macOS on Apple silicon (M5, 24 GB). The pilot runs on the CPU; the judge study, MedNLI fine-tuning and main study use the GPU through PyTorch Metal and MLX |
| License / use | Academic use. MTSamples-derived results are public. Outputs derived from PhysioNet data are never committed (results_private/ is git-ignored). |
| Thesis and deliverables | thesis/Ponangi_Thesis_Fall2026.pdf, built by thesis/build_thesis.py from the result files: 105 pages, 35 tables, 5 figures, 74 references; the progress slides and this description are built the same way (deliverables/). The built files are not tracked; one command regenerates each. |
| Reproducibility | Seeds: 42 (MTSamples sample, MedNLI fine-tuning, repeated-generation subset, selection-bias analyses), 1000r + k (repeated generations), 7 (audit sample) and 8 (its wave-2 extension), 0 (pilot bootstrap); pinned requirements and lock files; every pilot number is reproducible offline from results/claims_all.csv with recompute_metrics.py; main-study numbers require the credentialed data and are reproduced by run.sh --main-study |

## 2. File inventory

| Path | Size | Purpose |
|---|---|---|
| hallucination_eval.py | 27 KB | Main pipeline: loads MTSamples, samples 50 notes (seed 42), generates E0 (full note) and E1 (top-3 retrieved chunks only) summaries with GPT-4o-mini, segments claims (spaCy; header lines removed), retrieves top-3 evidence sentences (all-MiniLM-L6-v2), labels claims with cross-encoder/nli-MiniLM2-L6-H768, writes per-claim and per-summary results. |
| e2_extractive_eval.py | 20 KB | E2 extractive baseline: centroid ranking of source sentences by mean cosine similarity, top-5 kept in document order, no LLM call; evaluated with the same judge; appends E2 rows and columns to the result files. |
| e1b_fullnote_rag_eval.py | 4 KB | E1b: RAG with the full note plus the retrieved excerpts, and its evaluation. |

| Path | Size | Purpose |
|---|---|---|
| | | Requires OPENAI_API_KEY; 50 calls. |
| e3_cove_eval.py | 10 KB | E3: factored Chain-of-Verification on top of the E1 draft; per-claim verification against top-3 sentences from the full note, then revision (keep / replace / remove per claim); checkpointed with a --revise-only mode. Requires OPENAI_API_KEY; about 500 calls. |
| preprocessing.py | 4 KB | Shared utilities: is_markdown_header() removes section-header lines from the claim set; clean_mtsamples_text() and sentencize_cleaned() repair the comma-encoded line breaks of MTSamples. |
| recompute_metrics.py | 11 KB | Offline recomputation of every metric from results/claims_all.csv with headers excluded: per-sample UFR/CR, comparison table, aggregate statistics, Wilcoxon tests, bootstrap 95% CIs, effect sizes, header-filter effect tables. |
| ablations.py | 15 KB | Offline robustness analyses: NLI threshold sweep, evidence top-k (1, 3, 5), cleaned-evidence re-scoring, coverage proxy, negation analysis, keyword-assisted error taxonomy. |
| calibration_annptsumm.py | 12 KB | Judge validation against ann-pt-summ expert annotations (credentialed data, local models only): sentence-level precision/recall/F1/kappa, AUROC, threshold sweep, per-label recall, summary-level correlation. |
| calibration_variants.py | 7 KB | Five evidence-aggregation variants of the judge evaluated on the expert-annotated set. |
| judge_candidates.py | 14 KB | Judge selection study: six candidate judges (MiniLM, two DeBERTa-v3-large NLI models, MiniCheck DeBERTa and RoBERTa, Bespoke-MiniCheck-7B) plus the MedNLI-tuned model, each in two evidence modes (top-3 sentences; whole course in 400-token windows), scored against the 1,781 expert-annotated sentences with thresholds chosen on the other subset; writes results/judge_candidates.csv. |
| minicheck_server.py | 7 KB | Local HTTP scoring server for Bespoke-MiniCheck-7B (InternLM2.5-7B grounding checker, 8-bit MLX conversion kept outside the repository); returns the probability of 'Yes' for (document, claim) pairs with per-document prompt caching; queried by judge_candidates.py and mimic_evaluate.py over localhost. |
| finetune_mednli.py | 8 KB | Fine-tunes the strongest general NLI model on MedNLI (1 epoch, Adafactor, lr 2e-5, effective batch 16 as 4 x 4 accumulation, max length 128, embedding block frozen, MPS); saves the best-dev checkpoint outside the repository (~/Desktop/Thesis/models). |
| mimic_generate.py | 14 KB | Main-study generation against the local OpenAI-compatible mlx-lm server (Qwen2.5-7B-Instruct, 4-bit): E0, E1 (with citations) and its six retrieval variants, E1b, E2 and E3 for the 110 ann- |

| Path | Size | Purpose |
|---|---|---|
| | | pt-summ hospital courses; resumable JSONL output in results_private/. With the MIMIC_TAG environment variable, the same script runs the generator-robustness reruns (Meta-Llama-3.1-8B-Instruct-4bit, tag llama31; Mistral-7B-Instruct-v0.3-4bit, tag mistral7b) into tagged files; for a chat template that rejects a separate system role, the system prompt is folded into the user message. |
| mimic_extra.py | 6 KB | Main-study variants that separate confounds: E0 with a length instruction of about 150 words, E1b without the citation instruction, and two repeated generations of E0, E1, E1b and E3 on 40 courses (seeds 1000r + k); appended to the private generation file. |
| citations.py | 3 KB | Shared claim splitter: removes citation markers before spaCy segmentation, assigns each marker to its sentence by character position, and strips section headers merged into a sentence. |
| mimic_evaluate.py | 17 KB | Main-study evaluation with the judge of highest kappa: claim splitting (citations.py), forward UFR/CR, supported and unsupported counts, citation coverage, accuracy and reliability, reverse coverage of the clinician's discharge instructions and the number of their sentences not covered, scoring of the doctor-written instructions themselves, paired Wilcoxon tests with Holm adjustment over the primary family, ablation tests; aggregates to results/mimic_*.csv, per-claim rows to results_private/. |
| atomic_claims.py | 12 KB | Atomic-fact analysis: the local model splits every expert-set sentence and every main-study claim into atomic facts; each fact is scored by the selected judge; agreement with the experts and main-study rates at the level of facts (results/atomic_results.csv). |
| judge_selection_bias.py | 7 KB | Optimism and independence of the judge evaluation: nested cross-validation of the whole selection procedure by course, patient-disjoint transfer between the doctor-written and generated summaries, course-clustered bootstrap intervals and per-source agreement. |
| data_overlap.py | 5 KB | Matches every hospital course to its MIMIC-IV-Note discharge note by normalized text snippets and counts patient and admission overlap between the main study and the expert set (counts only). |
| mimic_revision_analysis.py | 16 KB | Variant tests, run-to-run variation, abstention analysis, coverage equivalence check, question-answering utility, per-condition bootstrap intervals, comparison of the first and the corrected claim splitter, and the overlapping-window check. |
| check_no_protected_text.py | 8 KB | Scans every tracked file, including PDF, Word and PowerPoint text, for eight-word |

| Path | Size | Purpose |
|---|---|---|
| | | sequences of the credentialed texts and of the private outputs; writes counts only. |
| mimic_second_judge.py | 6 KB | Second-judge agreement analysis: scores every base-condition claim with Bespoke-MiniCheck-7B (best mode and threshold from the judge study) and reports per-condition agreement, kappa, confirmed flags and UFR under both or either judge (results/mimic_judge_agreement.csv). |
| audit_sample.py, audit_tool.py, audit_map_types.py, audit_summary.py, audit_agreement.py | 11 KB + 15 KB + 3 KB + 9 KB + 3 KB | Blind manual audit of the judge in two passes (Section 7.9, protocol in Appendix C): course-clustered stratified sample and its wave-2 extension (60 courses, 340 claims, 60 citation checks; seeds 7 and 8), a local labeling page on 127.0.0.1 that hides the condition and the judge's label, keyword mapping of the annotator's free-text reasons to the protocol's problem types (reviewed in the tool), weighted estimates of the judge's precision, false omission rate, sensitivity, specificity and kappa and of the manually confirmed unsupported share, and the agreement between the two labeling passes (results/audit_*.csv). |
| verifact_eval.py | 23 KB | External check of the selected judge on MIMIC-III-Ext-VeriFact-BHC (PhysioNet, credentialed): scores the clinician-adjudicated propositions against the five best-matching 400-token windows of the summarized admission's notes (the admission's notes run to hundreds of windows) with the main study's windowing and threshold, checks the cap against all-window scoring on the shortest admissions, and computes agreement with the clinician labels and the published VeriFact raters' baseline (results/verifact_*.csv; runs locally, resumable). |
| mimic_qa.py | 7 KB | Question-answering pilot: three fixed questions per course answered from the full course (QA-E0) or three retrieved chunks (QA-E1) with an abstention instruction; evaluation of abstention rate, UFR/CR and support by the clinician's instructions. |
| analyze.py | 27 KB | Figures (results/fig_*.png, 200 dpi) and example tables (claims fixed by RAG, persistent unsupported claims). |
| thesis/build_thesis.py | 4 KB | Builds the thesis (.docx and .pdf) in CSUN format from the result files; content modules ch1_4.py, ch5_pilot.py, ch_judge.py, ch_mimic.py, ch8_9.py, appendices.py; two-pass table of contents, list of tables and list of figures via LibreOffice. |
| thesis/render_docx.py, thesis/results_loader.py, thesis/mimic_results.py, thesis/references.py | 27 KB + 8 KB + 9 KB + 21 KB | Rendering engine (python-docx), results access layers for the pilot and the main study, IEEE reference database with citation resolver. |
| deliverables/build_progress_slides.py | 37 KB | Progress report slides (PDF and PPTX). |
| deliverables/ build_source_data_description.py | 34 KB | This document. |

| Path | Size | Purpose |
|---|---|---|
| make_pdfs.py, build_thesis.py | 48 KB + 41 KB | Repository root: the May 2026 generators of the first progress report and the first thesis draft, superseded by thesis/build_thesis.py and the deliverables/ builders; kept for the record. |
| setup.sh, run.sh, requirements.txt, requirements-llm.txt | 2 KB + 4 KB + 0 KB + 0 KB | Environment bootstrap (setup.sh --llm also creates the model-server environment); reproduction chains (bash run.sh, bash run.sh --offline, bash run.sh --main-study); pinned dependencies, with complete lock files requirements-lock.txt and requirements-llm-lock.txt. |

## 3. Dependencies

requirements.txt (verbatim): # Pipeline environment (.venv, Python 3.9). Exact versions used for every reported result;; # requirements-lock.txt lists the complete environment including transitive dependencies.; openai==2.31.0; pandas==2.3.3; numpy==1.26.4; spacy==3.7.4; sentence-transformers==5.1.2; transformers==4.57.6; torch==2.8.0; scikit-learn==1.6.1; scipy==1.13.1; matplotlib==3.9.4; tqdm==4.67.3; python-docx==1.2.0; reportlab==4.4.10; python-pptx==1.0.2; rank-bm25==0.2.2; sentencepiece==0.2.2; protobuf==6.33.6; pillow==11.3.0.

Installed versions used for the reported results: openai 2.31.0, pandas 2.3.3, numpy 1.26.4, spacy 3.7.4 (en_core_web_sm), sentence-transformers 5.1.2, torch 2.8.0, scikit-learn 1.6.1, scipy 1.13.1, matplotlib 3.9.4, python-docx 1.2.0, reportlab 4.4.10, python-pptx 1.0.2. LibreOffice (soffice) converts .docx to PDF.

External models (downloaded once from the Hugging Face hub, then cached): sentence-transformers/all-MiniLM-L6-v2 (bi-encoder, 384-d, retrieval by cosine similarity); cross-encoder/nli-MiniLM2-L6-H768 (NLI classifier; outputs contradiction/entailment/neutral). External API: OpenAI GPT-4o-mini, temperature 0.3, max 450 output tokens; key supplied via the OPENAI_API_KEY environment variable and never stored in the repository. Judge study and main study: MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli, cross-encoder/nli-deberta-v3-large, lytang/MiniCheck-DeBERTa-v3-Large and lytang/MiniCheck-RoBERTa-Large (candidate judges); mlx-community/Qwen2.5-7B-Instruct-4bit served with mlx-lm 0.31 (mlx 0.32) through an OpenAI-compatible interface on localhost; bespokelabs/Bespoke-MiniCheck-7B converted to 8-bit MLX and served by minicheck_server.py (judge candidate); rank-bm25 for the BM25 ablation. No protected text is sent to any external service.

## 4. Data sources

| Source | Type | Content | Size | Access status and location |
|---|---|---|---|---|
| MTSamples | Public CSV | 4,999 sample medical transcriptions, 40 specialties; 624 eligible Discharge Summary and Consult - History and Physical notes; 50 sampled (seed 42): 40 consultations, 10 discharge summaries; source length 117 to 1,809 words (mean 539); summaries generated under five conditions | 17 MB | Public. Stored one directory above the repository (../mtsamples.csv), not tracked. |
| MIMIC-IV-Note v2.2 | Credentialed CSV | 331,794 de-identified | 1.8 GB compressed | Credentialing approved |

| Source | Type | Content | Size | Access status and location |
|---|---|---|---|---|
| | (gzip) | discharge summaries and about 2.3 million radiology reports (PhysioNet doi 10.13026/1n74-ne17) | | 4/28/2026; downloaded 5/14/2026; gzip integrity verified; SHA-256 manifest present. Stored locally (~/Desktop/Thesis). Used only to match each hospital course to its discharge note for the patient-overlap check (data_overlap.py); not used for generation. |
| ann-pt-summ v1.0.1 | Credentialed JSONL/XML | Main-study corpus (110 MIMIC-IV hospital courses paired with the discharge instructions their clinicians wrote) and expert span annotations of unsupported facts: 100 doctor-written and 100 LLM-generated patient summaries plus 10 validation summaries; 423 agreed spans in 10 label types (PhysioNet doi 10.13026/gedc-j464) | 2.3 MB used (archive incomplete: 934 MB of >3.4 GB; the 14 annotation and derived files verified against SHA-256) | DUA accepted; downloaded 5/12/2026. Used for the judge validation and selection and as the main-study corpus, with local models only. Generations, claims and labels in results_private/ (git-ignored). |
| MIMIC-III-Ext-VeriFact-BHC v1.0.0 | Credentialed CSV (gzip) | 13,070 propositions decomposed from human-written and LLM-written Brief Hospital Course summaries of 100 MIMIC-III patients, each adjudicated by clinicians as supported, not supported or not addressed (12,614 valid); the patients' 4,787 EHR notes; and the verdicts of 72 configurations of the dataset's own AI raters (Chung et al., NEJM AI 2026, doi 10.1056/AIdbp2500418) | 400 MB extracted (419 MB archive) | DUA accepted; downloaded 11/1/2026; the 14 files SHA-256-verified; stored locally (~/Desktop/Thesis/verifact-bhc). Used only for the external check of the judge (Section 6.7) with local models; per-proposition scores in results_private/ (git-ignored). |
| MedNLI v1.0.0 | Credentialed JSONL | 14,049 clinician-written premise and hypothesis pairs from MIMIC-III notes (11,232 train, 1,395 dev, 1,422 test; entailment, neutral, contradiction) used to fine-tune the judge (PhysioNet doi 10.13026/C2RS98) | 9 MB | Data use agreement accepted and downloaded 9/15/2026; stored locally (~/Desktop/Thesis/mednli); never committed |

Compliance. CITI Program courses 'Data or Specimens Only Research' and 'Conflicts of Interest' completed April 10, 2026 (records 76432215 and 76432216). PhysioNet credentialed data and any output

containing text derived from them are never committed, never uploaded to shared cloud storage, and never transmitted to the OpenAI API. Only MTSamples-based generations were sent to the API.

## 5. Result files (results/)

| File | Content |
|---|---|
| claims_all.csv | 2,220 rows: one per claim and condition; columns doc_id, condition, specialty, description, claim, evidence (top-3 sentences, '\|'-separated), evidence_scores, label, p_entailment, p_contradiction, is_header. 162 rows are header lines (is_header = True) and are excluded from all metrics; 2,058 claims remain ({'E0': 522, 'E1': 447, 'E1b': 510, 'E2': 242, 'E3': 337}). |
| summaries.csv | 50 rows: source note metadata, the E0/E1/E2 summary texts and word counts, and per-condition UFR, CR and claim counts. |
| comparison_per_sample.csv | 50 rows: per-document UFR, CR and claim counts for every condition, plus paired deltas. |
| aggregate_statistics.csv | Mean, SD, median per condition; E1−E0 and E2−E0 deltas, % improved, Wilcoxon p, bootstrap 95% CI. |
| pairwise_tests.csv | Every condition pair × metric: means, medians, mean and median delta, bootstrap CI, W, p, d_z, rank-biserial, % improved/unchanged/worse. |
| header_filter_effect.csv, header_filter_tests.csv | Metrics and E1-vs-E0 tests before and after removing header lines. |
| ablation_thresholds.csv, ablation_topk.csv, ablation_cleaned_evidence.csv, claims_cleaned_evidence.csv | Robustness analyses of the pilot (Section 5.4). |
| coverage_per_sample.csv, coverage_summary.csv | Coverage proxy per document and per condition. |
| e2_error_analysis.csv, error_taxonomy_counts.csv, error_taxonomy_examples.csv | Negation analysis and error taxonomy. |
| calibration_overall.csv, calibration_by_label.csv, calibration_label_counts.csv, calibration_threshold_sweep.csv, calibration_summary_level.csv, calibration_by_system_means.csv, calibration_variants.csv | Judge validation against expert annotations (Section 5.3; aggregate statistics only, no MIMIC-derived text). |
| examples_fixed_by_rag.csv, examples_persistent.csv | Qualitative example pairs. |
| judge_candidates.csv | Judge selection study: AUROC, threshold, flag rate, precision, recall, specificity, F1 and kappa for every candidate judge, evidence mode and sentence subset (Chapter 6). |
| mimic_summary.csv, mimic_pairwise_tests.csv, mimic_reference.csv | Main study aggregates: per-condition means and medians of UFR, CR, coverage of the clinician's instructions, citation accuracy, words, claims and abstentions; paired tests against E0, E1 and the doctor-written instructions; the judge's rates on the doctor-written instructions next to the expert span counts (Chapter 7). No MIMIC-derived text. |
| mimic_judge_agreement.csv, mimic_second_judge_tests.csv | Agreement between the selected judge and Bespoke-MiniCheck-7B on the 6,588 base-condition claims, per condition, and paired Wilcoxon tests of per-document UFR under the second judge and under both judges (Section 7.10). |
| mimic_ablations.csv, mimic_qa_summary.csv, mimic_qa_utility.csv | Retrieval ablations (paired differences of six E1 variants from the base configuration) and the question-answering pilot aggregates by condition and question type, including the shares of all questions abstained on, answered fully supported and answered with an unsupported sentence. |
| mimic_condition_ci.csv, mimic_variant_tests.csv, mimic_replicates.csv, mimic_abstention.csv, mimic_equivalence.csv, mimic_window_check.csv, mimic_splitting_check.csv, mimic_tau05/ | Revision analyses: per-condition bootstrap intervals, length-matched and no-citation variants, run-to-run variation, abstention, coverage equivalence check, overlapping-window check, splitter comparison, and the main-study aggregates at threshold 0.5. |
| atomic_results.csv, judge_selection_bias.csv, judge_selection_frequency.csv, judge_per_source.csv, data_overlap.csv | Atomic-fact analysis (Sections 6.6, 7.10), optimism and transfer of the judge selection and per-source agreement (Section 6.5), and patient and admission overlap (counts only). |
| audit_summary.csv, audit_types.csv, audit_citations.csv, audit_agreement.csv, audit_*_v1.csv, protected_text_scan.csv, protected_text_scan_summary.csv | Manual audit aggregates for the second labeling pass (Section 7.9), the first pass's aggregates under _v1 names, agreement between the passes, and the result of the protected-text scan (Section 3.7). |
| verifact_judge_eval.csv, verifact_retriever_check.csv, | External check of the judge on VeriFact-BHC (Section 6.7): |

| File | Content |
|---|---|
| verifact_ai_raters.csv | agreement with the clinician labels overall and by author, proposition type and label; the cap check against all-window scoring; the window-ranking comparison on the shortest admissions; and the agreement of the dataset's own 72 AI-rater configurations. No MIMIC-derived text. |
| mimic_summary_llama31.csv, mimic_pairwise_tests_llama31.csv, mimic_reference_llama31.csv, and the same for mistral7b | Generator-robustness aggregates: the main-study measures and paired tests recomputed on summaries generated by Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 (Section 7.10 of the thesis). No MIMIC-derived text. |
| fig_*.png | 19 figures at 200 dpi used in the thesis. |

## 6. Pipeline architecture

- Stage 1, generation: E0 prompts GPT-4o-mini with the first 4,500 characters of the note; E1 prompts it with the top-3 retrieved five-sentence chunks only (query = description + first 300 characters; all-MiniLM-L6-v2 cosine; at most 4,000 characters); E1b adds the full note to the same excerpts; E3 verifies each E1 claim against the top-3 sentences retrieved from the full note and rewrites the summary; E2 selects the top-5 centroid sentences without an LLM.
- Stage 2, claim segmentation: spaCy en_core_web_sm sentences of at least 10 characters; lines that are only a markdown/section header are removed (preprocessing.is_markdown_header).
- Stage 3, evidence retrieval: for each claim the 3 most similar source sentences by cosine similarity of all-MiniLM-L6-v2 embeddings.
- Stage 4, NLI classification: cross-encoder/nli-MiniLM2-L6-H768 scores each (evidence sentence, claim) pair; the claim's p_entailment and p_contradiction are the maxima over the three pairs. Supported if p_entailment ≥ 0.5 and p_entailment > p_contradiction; Contradicted if p_contradiction ≥ 0.5 and p_contradiction > p_entailment; otherwise Not-Supported.
- Stage 5, aggregation: per summary, UFR = (Contradicted + Not-Supported) / claims and CR = Contradicted / claims.
- Stage 6, statistics: two-sided Wilcoxon signed-rank tests on paired per-document metrics, percentile bootstrap 95% CIs (10,000 resamples), d_z and rank-biserial effect sizes, Holm adjustment over all twenty pilot pairwise tests (thesis/results_loader.py); in the main study, over the twelve primary tests of E1, E1b, E2 and E3 against E0 on UFR, CR and coverage (mimic_evaluate.py) and within each measure for the ablations.
- Validation: the unchanged judge is applied to 210 expert-annotated summaries; sentence-level agreement with expert spans, threshold sweep, AUROC, per-label recall, summary-level Spearman correlation, and five evidence-aggregation variants (calibration_annptsumm.py, calibration_variants.py).
- Judge selection: six candidate judges in two evidence modes are scored on the same 1,781 sentences with thresholds chosen on the other subset; the strongest general model is fine-tuned on MedNLI and scored again; the candidate with the highest kappa becomes the main-study judge (judge_candidates.py, finetune_mednli.py).
- Main study, generation: Qwen2.5-7B-Instruct (4-bit, mlx-lm, localhost) writes four-part patient summaries of 110 MIMIC-IV hospital courses under E0, E1 (five-sentence chunks, top-3 dense retrieval, citations required), E1b, E2 and E3, plus six E1 variants (3- or 8-sentence chunks, top-2 or top-5, BM25, no citations); the retrieval query is a fixed section list plus the first 300 characters (mimic_generate.py).
- Main study, evaluation: the selected judge scores every claim against the whole course in windows (UFR, CR), cited excerpts (citation accuracy) and, in the reverse direction, every sentence of the clinician's discharge instructions against the summary (coverage); the doctor-written instructions are scored as a human reference; paired Wilcoxon tests against E0, E1 and the reference, and against the E1 base configuration for the ablations (mimic_evaluate.py).
- Question-answering pilot: three questions per course (medications, follow-up, warning signs) answered from the full course or from three retrieved chunks with an abstention instruction; abstention rate, UFR/CR of answered questions, and support of the answers by the clinician's instructions (mimic_qa.py).

- Manual audit: 340 claims and 60 citation checks sampled by course and stratum from E0, E1, E1b and E3 and labeled blind in a local tool, in two passes under the protocol of Appendix C; stratum-weighted error rates of the judge, manually confirmed unsupported shares per condition, and the agreement between the passes (audit_sample.py, audit_tool.py, audit_map_types.py, audit_summary.py, audit_agreement.py).
- External check: the selected judge, unchanged, scores the 12,614 clinician-adjudicated VeriFact-BHC propositions against the five best-matching 400-token windows of the admission's notes (dense and BM25 rankings interleaved); the cap is checked against all-window scoring on the shortest admissions and the dataset's published AI raters serve as a baseline (verifact_eval.py).

## 7. How to reproduce

```
git clone https://github.com/slay-a/medical-hallucination-eval.git && cd medical-hallucination-eval
bash setup.sh                        # creates .venv, installs pinned dependencies, downloads en_core_web_sm
# place mtsamples.csv one directory above the repository
export OPENAI_API_KEY='sk-...'       # needed only for generation (E0, E1; optionally E1b, E3)
bash run.sh                          # generation + evaluation + recomputation + ablations + figures
bash run.sh --offline                # reproduce every pilot number from the stored results without any API call
bash run.sh --main-study             # judge study, MedNLI fine-tuning, MIMIC-IV main study and QA pilot (credentialed data, local models)
python thesis/build_thesis.py        # rebuild the thesis .docx and .pdf
python verifact_eval.py --baseline   # published VeriFact raters against the clinician labels
python verifact_eval.py --score --premise admission --batch 8 --chunk 256   # external check of the judge (resumable, about a day)
python verifact_eval.py --summarize --premise admission                     # metrics -> results/verifact_judge_eval.csv
python deliverables/build_progress_slides.py && python deliverables/build_source_data_description.py
```

## 8. Changes between versions of this description

| Versions | What changed |
|---|---|
| May to September 4, 2026 | Header lines excluded from the claim set (162 of 1,373 stored E0/E1/E2 claim rows), which changes every metric; the E1 description corrected (excerpts only, non-overlapping five-sentence chunks); E1b and E3 generated and evaluated; four imported dependencies added to requirements.txt; bootstrap intervals, effect sizes and Holm adjustment computed; the ann-pt-summ DUA signed and the annotation files in use. |
| September 4 to October 12, 2026 | A judge selection study replaced the pilot judge with a DeBERTa-v3-large NLI model validated against the experts; a MedNLI fine-tuning script; the main study on 110 MIMIC-IV hospital courses with a locally served Qwen2.5-7B-Instruct (five conditions with citations, coverage against the clinician's instructions, citation accuracy, six retrieval ablations, question-answering pilot). |
| October 12 to November 1, 2026 | Citation markers removed before sentence segmentation and assigned by position, merged section headers stripped (citations.py), which corrects the citation measures; length-matched and no-citation variants, repeated generations, atomic-fact scoring, a selection-bias analysis, a patient-overlap check, an overlapping-window check, per-condition intervals and the blind manual-audit tool; dependencies pinned with lock files; a scan |

| Versions | What changed |
|---|---|
| | confirming that no tracked file contains protected text. |
| November 1 to November 9, 2026 | The manual audit relabeled in a second blind pass under a clarified rubric on an extended sample (400 items), with the agreement between the passes (audit_agreement.py) and the problem types mapped from the annotator's notes (audit_map_types.py); the four main-study conditions regenerated and re-evaluated with Llama-3.1-8B-Instruct and Mistral-7B-Instruct-v0.3 (MIMIC_TAG in mimic_generate.py and mimic_evaluate.py); verifact_eval.py added and run for the external check of the judge (Section 6.7). |
| November 9 to November 10, 2026 | No result file or number changed. The thesis text modules were rewritten in a compact form (105 pages from 163): each result is a table or figure with a short paragraph, the methods state what was done, and the discussion interprets without repeating the results; the per-document results, worked examples and threshold-sweep appendices were removed, so the code appendix is now Appendix B and the audit protocol Appendix C; the progress slides and this description were rebuilt to match. |